Leiden Community Detection

Student Social Network — Neo4j GDS ·

1. Objective

This document presents the results of applying the Leiden algorithm to the student friendship network. Leiden is an improvement over Louvain that guarantees well-connected communities — it adds a refinement phase that prevents Louvain's known issue of producing internally disconnected communities. It also uses a randomized local move phase that allows it to escape local optima, generally producing higher modularity in more iterations.

2. Implementation

Graph Projection

CALL gds.graph.project('student-friendship-graph-leiden', 'Student', { EST_AMI_DE: { orientation: 'UNDIRECTED' } }) YIELD graphName, nodeCount, relationshipCount;

text

Leiden Write

CALL gds.leiden.write('student-friendship-graph-leiden', { writeProperty: 'leidenCommunity' }) YIELD communityCount, modularity, modularities, nodePropertiesWritten;

text

3. Global Results

Metric	Value
Number of communities	15
Final modularity Q	0.3240
Nodes written	1,500

Metric	Value
Iterations	4
Modularity — iteration 1	0.2376
Modularity — iteration 2	0.3091
Modularity — iteration 3	0.3220
Modularity — iteration 4	0.3240

Modularity Progression:
Iteration 1 → 2 → 3 → 4
0. 2376 → 0.3091 → 0.3220 → 0.3240
(threshold Q=0.30 indicated)

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4. Community Size Analysis

Leiden produced 15 communities ordered by decreasing size. The two largest communities (4 and 5) contain 199 and 197 students respectively, together representing 26% of the entire network. The smallest community (6) has only 36 students. Community IDs in Leiden are sequential integers starting from 1.

Community Sizes:

Community ID	Size	% of total
4	199	13.3%
5	197	13.1%
26	143	9.5%
13	139	9.3%
31	120	8.0%
16	108	7.2%
29	103	6.9%
1	95	6.3%

Community ID	Size	% of total
2	81	5.4%
18	72	4.8%
10	68	4.5%
19	55	3.7%
36	47	3.1%
15	37	2.5%
6	36	2.4%

5. Field of Study Distribution per Community

The stacked chart below shows the composition of each community by field of study. No single department dominates any community, confirming that social connections in this network cross departmental boundaries freely.

Distribution for the top 6 largest communities:

Community	Software Eng.	Electrical Eng.	Mechanical Eng.	Industrial Eng.	Networking Eng.	Other	Total
4	45	25	23	23	19	64	199
5	44	20	24	30	18	61	197
26	16	20	32	16	21	38	143
13	31	14	18	18	21	37	139
31	21	22	17	11	13	36	120
16	27	16	14	12	6	33	108

6. Observations & Analysis

- **Modularity above threshold**
  The final modularity Q = 0.3240 exceeds the 0.30 threshold considered significant in the literature. This confirms that the Leiden algorithm has detected a genuine community structure in the student friendship network.

- **Convergence in 4 iterations**
  The algorithm converged in 4 iterations, going from Q=0.2376 to Q=0.3091 to Q=0.3220 to Q=0.3240. The largest gain happened between iteration 1 and 2 (+0.0715), with diminishing returns in iterations 3 and 4, indicating stable convergence.

- **Sequential community IDs**
  Leiden assigns sequential integers starting from 1 as community IDs. This makes the results significantly easier to read, display, and work with in the recommendation system compared to algorithms that use internal node IDs.

- **Communities are not field-based**
  Every community contains students from all 6 fields of study in mixed proportions. The algorithm grouped students by social connections, not by academic department. Friendships in this network cross departmental boundaries freely.

- **High size variance**
  Community sizes range from 36 (community 6) to 199 (community 4). Two communities — 4 and 5 — together account for 26% of all students, suggesting a core-periphery structure where a few large social hubs anchor the biggest communities.

- **Well-connectivity guaranteed**
  Leiden's key theoretical advantage is that every detected community is guaranteed to be internally well-connected. This property makes the partitioning more reliable for downstream tasks such as recommendation and community profiling.

- **Recommendation system implication**
  Since communities are socially formed and not field-based, the `leidenCommunity` property written to each Student node can be safely used as a feature in the recommendation system without inadvertently creating field-based filter bubbles.

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7. Conclusion

The Leiden algorithm successfully partitioned the 1,500-student friendship network into 15 well-connected communities with a final modularity of 0.3240. The communities reflect genuine social structures rather than demographic or academic groupings. The well-connectivity guarantee and sequential community IDs make Leiden a strong candidate for integration into the recommendation system. .